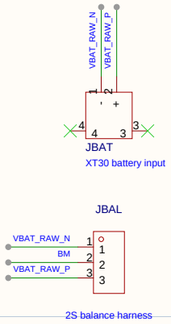
DesktopAssistant Integrated | Battery protection and balance / 电池保护与均衡

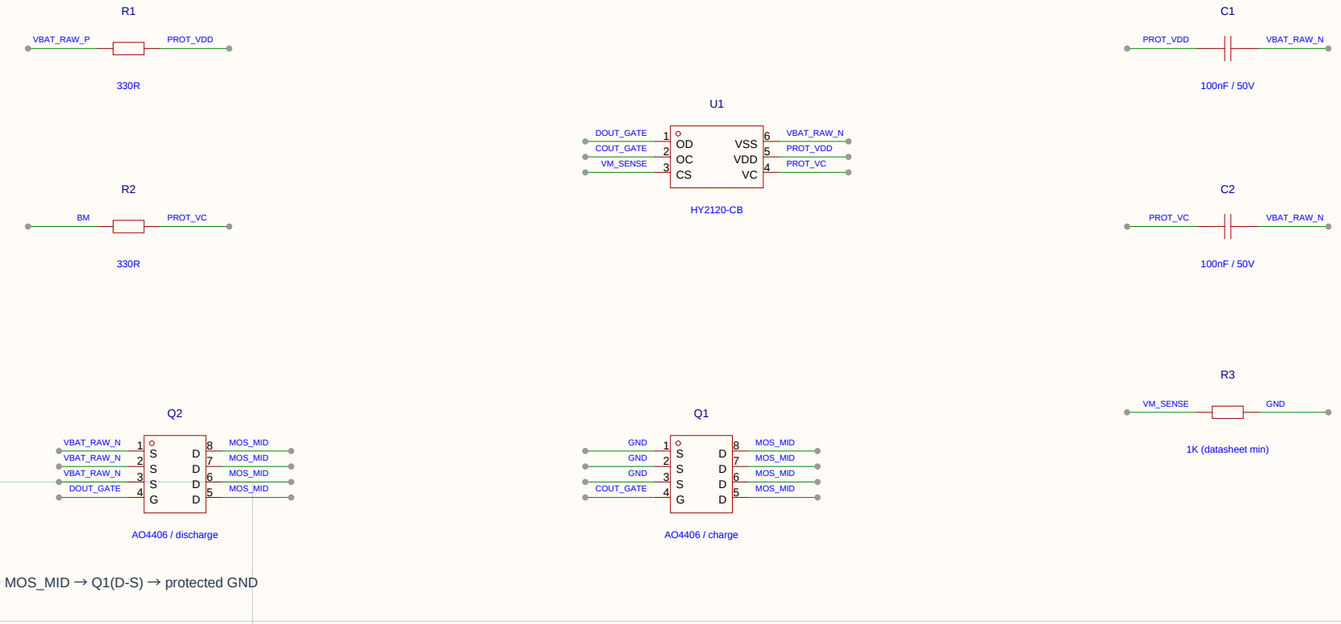
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2S battery input / 原始电芯端



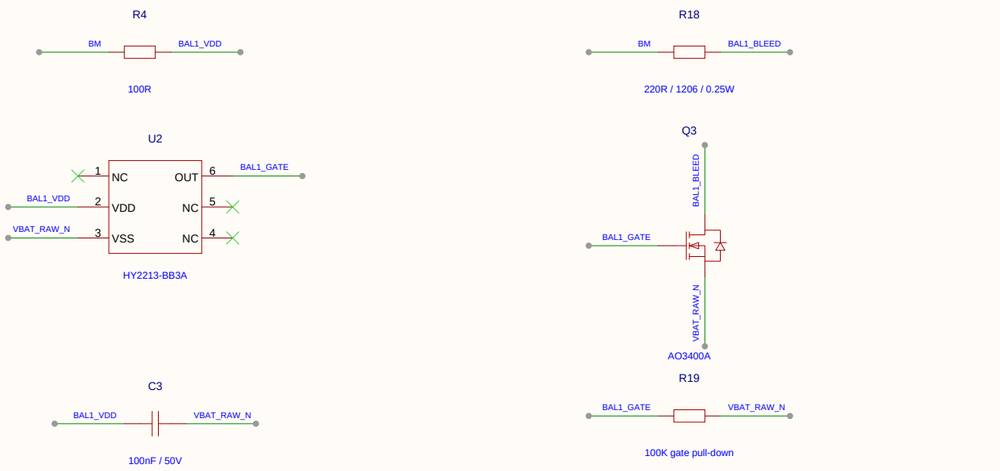
JBAL: 1=B , 2=BM, 3=B+; verify cable polarity.

HY2120-CB protection / 过充过放与短路保护

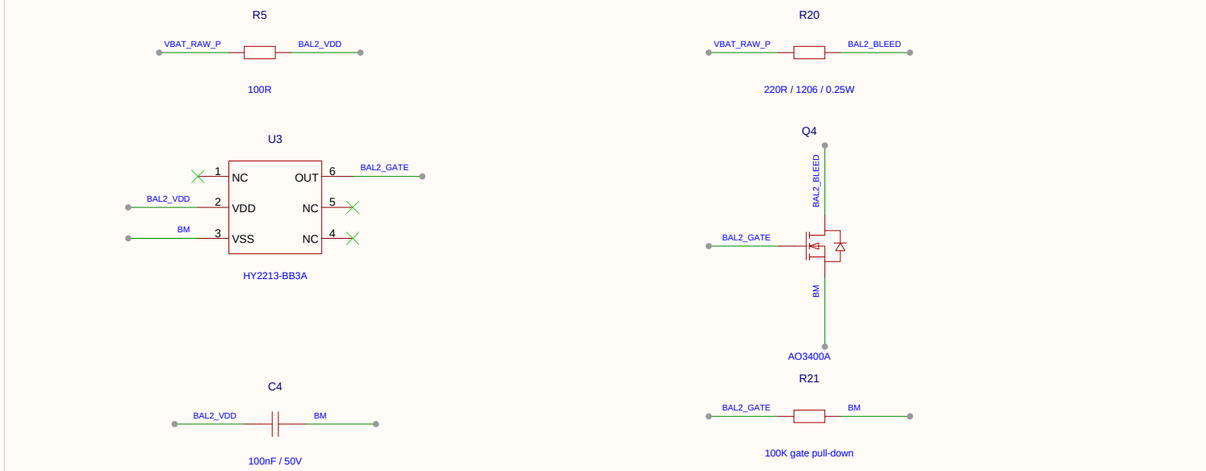


B → Q2(S-D) → MOS_MID → Q1(D-S) → protected GND

Lower cell balance / 下节电芯



Upper cell balance / 上节电芯



Balance current ≈ 19mA at 4.2V; each bleed resistor ≈ 0.08W.

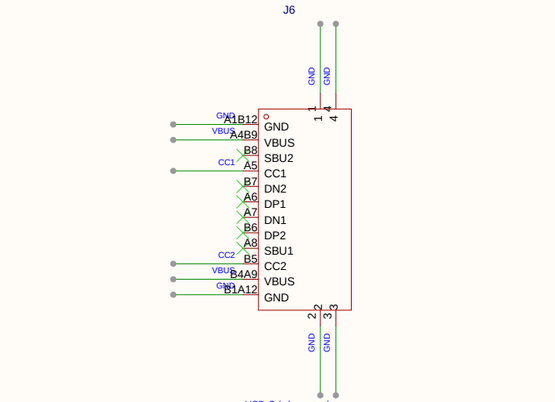
GND is protected pack negative. Do not short GND to VBAT_RAW_N.

Protection and balance thresholds require bench validation before battery use.

DesktopAssistant Integrated | USB-C charger / 2S 锂电充电

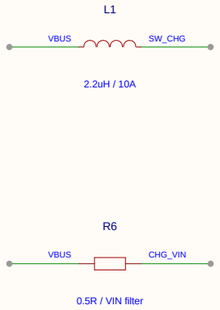
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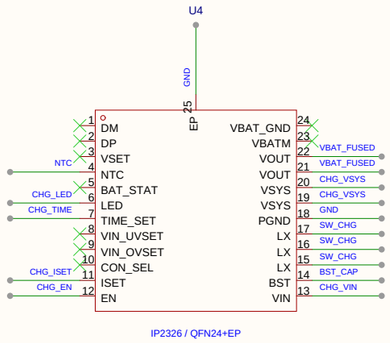
5V USB-C input / 专用 5V 适配器

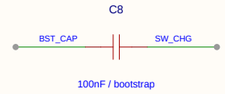


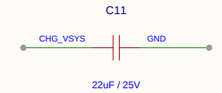


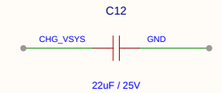
IP2326 synchronous boost charger



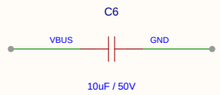


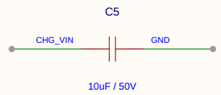


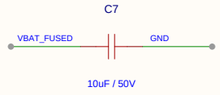


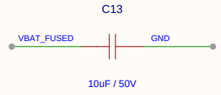


Input / output decoupling

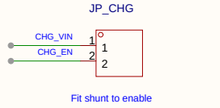






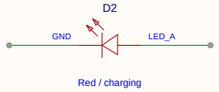


Charge enable and status / 默认禁止充电

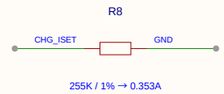


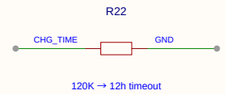


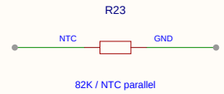


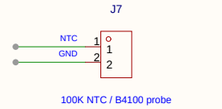


Charge settings / 电流、超时和电池温度









2S: CON_SEL NC; 8.4V: VSET NC; VIN UV/OV: NC defaults.

External HY2213 balance: IP2326 pins 23/24 left NC.

Use regulated 5V ≥ 2A adapter. D+/D unused; no QC request.

JP_CHG defaults OPEN. NTC must touch battery; charge with main switch OFF.

0.353A is calculated; validate charge current/termination before battery use.

DesktopAssistant Integrated | Power and MCU / 配电与主控

Revision 1.1 2026-09-12 03 / 05

Protected pack → fuse → switch → regulators

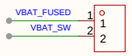
F1



01530008Z fuse holder

Fit 029707.5WXNV 7.5A fuse into F1.

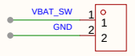
J_SW



External DC power switch

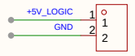
1=fused input; 2=switched output.

J_LOG_IN



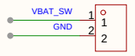
MP1584 module VIN ±

J_LOG_OUT



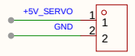
MP1584 module VOUT ±

J_SRV_IN



DFR0753 module VIN ±

J_SRV_OUT

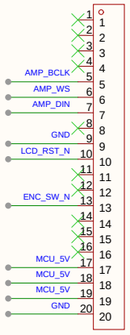


DFR0753 module VOUT ±

Two external regulator modules; outputs are separate. Set both to 5.0V before connection.

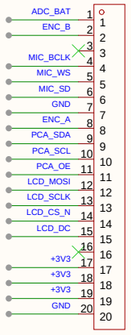
ESP32-S3 carrier sockets / 保留已核对的开发板针序

J1



ESP32 right row R1-R20

J2

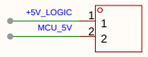


ESP32 left row L1-L20

MCU 5V isolated by JP_MCU; OPEN during USB debugging.

Logic supply / USB isolation

JP_MCU



MCU supply shunt / OPEN

C10



470uF / 16V

Battery ADC / switched measurement

R10



100K / 1%

R9



33K / 1%

C14



100nF low-pass

ADC_BAT ≈ 2.08V at pack 8.4V.

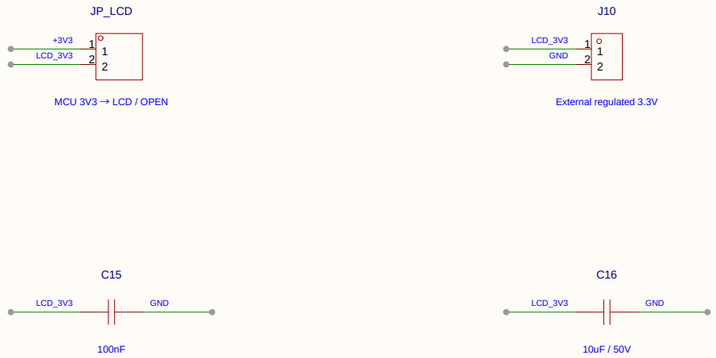
Each module connector: pin 1 positive, pin 2 GND. Modules are external, not on-board footprints.

J_SW is a cable interface, not a diode. Use a DC-rated switch appropriate for servo startup current.

DesktopAssistant Integrated | Display, audio and controls / 显示音频与交互

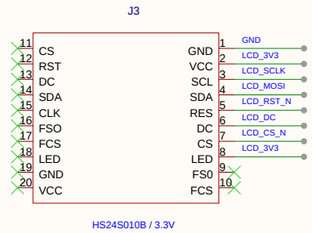
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3.3V LCD supply / 独立隔离的屏幕支路



Select ONE LCD source. Keep JP_LCD open when J10 is used.

SPI display / HS24S010B



Pins 1-10 = primary interface; alternate pins 11-20 unused.

I2S module harnesses / 音频线束接口



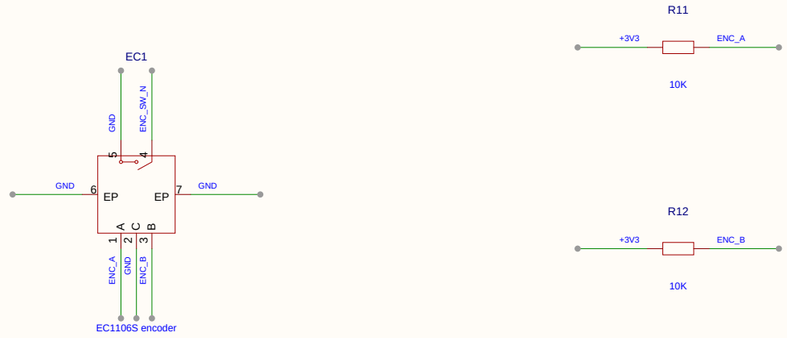
J4: 1=BCLK, 2=WS, 3=SD, 4=3V3, 5=GND, 6=L/R GND.

J5: 1=BCLK, 2=WS, 3=DIN, 4=5V, 5=GND, 6=GND aux.

These are wire-harness pinouts; follow signal names on the modules.

Speaker connects to amplifier OUT+/OUT , not motherboard GND.

EC11 controls / 旋钮与按压



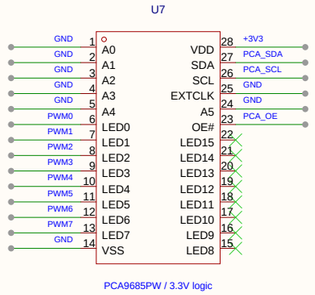
All ESP32 signals are 3.3V. Debounce encoder and button in firmware.

LCD source follows RevB isolation strategy. This page does not connect the two regulator outputs.

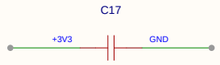
DesktopAssistant Integrated | PWM and servos / 舵机控制与配电

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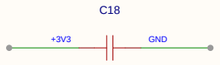
PCA9685 " I2C 0x40 " OE high = outputs disabled



PCA9685PW / 3.3V logic



100nF near VDD



10uF local bypass

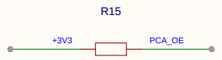
I2C pull-ups and safe startup



10K SDA

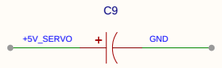


10K SCL



10K OE pull-up

OE defaults HIGH. Firmware must initialize PWM before enabling.

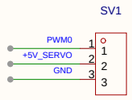


1000uF / 10V bulk

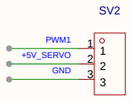


100nF ceramic

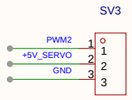
Eight 3-pin servo headers / 1=PWM, 2=+5V_SERVO, 3=GND



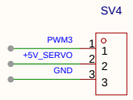
Servo 0



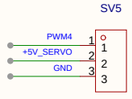
Servo 1



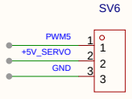
Servo 2



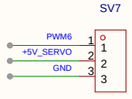
Servo 3



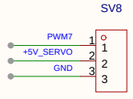
Servo 4



Servo 5



Servo 6



Servo 7

Servo power has its own 5V converter; route high-current return directly to protected power ground.

LED8-LED15 unused. EXTCLK tied to GND for internal oscillator. All address pins are GND.

No servo motion until supply, polarity, current limit and firmware fail-safe behavior have been tested.